KYLE BINDER

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APPOINTMENTS

Louisiana State University 2025 – Present
RTG Postdoctoral Researcher

EDUCATION

The Ohio State University 2019 – 2025
M.S. & Ph.D. in Mathematics *Advisor: Eric Katz*

Baylor University 2015 – 2019
B.A. in Mathematics and Linguistics (University Scholars)

RESEARCH INTERESTS

Combinatorial algebraic geometry, combinatorics, toric varieties, matroids, tropical geometry.

PAPERS AND PREPRINTS

1. *Tropical Toric Vector Bundles on Projective Space* (with Sergio Cristancho, Caitlin Davis, Christopher Manon, Emma Pickard, Mahrud Sayrafi), in preparation.
(∼ 25pp., expected December 2026)

2. *Representations on the Top-Nonvanishing Weight Cohomology of a Smooth Toric Variety*, in preparation.
(∼20 pp., expected November 2026)

3. *Augmented Singular Cohomology, Uniform Matroids, and Real-Rootedness* (with Lorenzo Vecchi), submitted, arXiv:2609.15946. 27 pp.

4. *The Singular Cohomology Ring of a Uniform Matroid: Combinatorics and Lefschetz Properties*, submitted, arXiv:2606.07490. 21 pp.

5. *The Singular Cohomology Ring of a Matroid*, submitted, arXiv:2412.05732. 35 pp.

6. *The Unipotent Tropical Fundamental Group* (with Eric Katz), *J. London Math. Soc.* (2026), 114:e70701. https://doi.org/10.1112/jlms.70701. 55 pp.

7. *Limit Sets and Internal Transitivity in Free Group Actions* (with Jonathan Meddaugh), preprint, arXiv:1908.07382. 17 pp.

8. *On 2-coloring the integers to forbid a distance and monochromatic translates of a triangle*, *Geombinatorics* **27** (2018), no. 4, 137–146.

TALKS

The Singular Cohomology Ring of a Matroid November 2026
University of Toronto Combinatorics Seminar

The Singular Cohomology Ring of a Matroid November 2026
Brown Combinatorics Seminar

The Singular Cohomology Ring of a Matroid September 2026
Combinatorics & Geometry BLT Seminar

A Tropical Approach to the Unipotent Fundamental Group of a Hyperplane Arrangement Complement — October 2025
Hyperplane Arrangements and Polytopes Special Session
AMS Southeastern Sectional Meeting

Cohomology of Toric Varieties from Matroids — September 2025
LSU Geometry and Topology Seminar

The Tropical Unipotent Fundamental Group — October 2024
5 Minute Talk at Tropical Geometry: Moduli Spaces and Matroids Workshop

Higher Tor Groups of Matroids — May 2024
5 Minute Talk at Workshop of the Kentucky–Ohio Algebra Alliance (KOALA)

Higher Tor Groups of Matroids — April 2024
OSU Combinatorics and Probability Seminar

Higher Tor Groups of Stanley–Reisner Rings of Matroids — May 2023
5 Minute Talk at Workshop of the Kentucky–Ohio Algebra Alliance (KOALA)

The Tropical Unipotent Fundamental Group — January 2023
20 Minute Talk at Workshop on Matroids and Tropical Combinatorics (MaTroCom)

A Survey of Tropical Cohomology — September 2022
OSU Algebraic Geometry Seminar

The Geometries of Hyperplane Arrangement Complements — April 2022
OSU Student Algebraic Geometry Seminar

p-adic Integration on Hyperelliptic Curves — February 2021
OSU Student Algebraic Geometry Seminar

Posters

The Singular Cohomology Ring of a Matroid — July 2026
Combinatorics at the Confluence

The Singular Cohomology Ring of a Matroid — April 2026
Texas Algebraic Geometry Symposium

The Singular Cohomology Ring of a Matroid — December 2025
Arizona–New Mexico Symposium on Commutative Algebra

What is the Cohomology of a Non-Compact Toric Variety? — December 2023
Kentucky, Ohio, Indiana (KOI) Combinatorics Lectures

Teaching

LSU Math 4023 – Applied Algebra — (Expected) Spring 2027
Instructor

LSU Math 2065 – Elementary Differential Equations — (Expected) Spring 2027
Instructor

LSU Math 4997 – Vertically Integrated Research: Combinatorial Topology — Fall 2026
Instructor

LSU Math 1550 – Calculus 1 — Fall 2025
Instructor

OSU MATH 1131 – Business Calculus — Spring 2024
Recitation Instructor

OSU MATH 1131 – Business Calculus — Autumn 2022
Recitation Instructor

OSU MATH 1148 – College Algebra — Autumn 2021
Recitation Instructor

OSU MATH 1148 – College Algebra — Autumn 2020
Recitation Instructor, Virtual (Zoom)

MENTORSHIP

Mentor for Undergraduate/Graduate Research Project — Fall 2026
Project on h-vectors of matroids (2 undergraduate/2 graduate students)

Directed Reading Program (DRP) Mentor — 2021 – Present

1. "The Mathematics of Chip-Firing" with E.R. (Fall 2026)
2. "The Mathematics of Chip-Firing" with C.B. (Fall 2023)
3. "Combinatorial Optimization" with G.M. (Spring 2023)
4. "An Invitation to Applied Category Theory" with J.W. (Spring 2022)
5. "Quaternions for Computer Graphics" with S.B. (Fall 2021)
6. "Basic Concepts of Algebraic Topology" with D.L. (Spring 2021)

SERVICE AND ORGANIZATION

Special Session on Algebraic and Geometric Aspects of Matroid Theory at AMS Central Sectional Meeting — October 2026
Co-organizer

LSU Geometry and Topology Seminar — 2025 – Present
Co-organizer

OSU Directed Reading Program (DRP) — 2021 – 2025
Co-organizer

Kentucky, Ohio, Indiana (KOI) Combinatorics Lectures — December 2023
Local Organizer

HONORS AND AWARDS

OSU Distinguished University Fellowship — 2019 – 2020, 2024 – 2025

Tibor Radó Research Semester — Spring 2021

Phi Beta Kappa — Fall 2018